20 EXAM PRACTICE

1 Organic solids are often purified by recrystallisation. What is the basis for this technique?

A the impurities must be insoluble in the solvent used
B the impurities must react with the solvent used
C the impurities crystallise first when the hot solution is cooled
D the cooled solution is saturated with the desired material but not with the impurities. [1]

(Total for Question 1 = 1 mark)

2 Steam distillation may be used in the purification of some compounds. What does this technique depend on?

A the compound being purified forms a single layer with water
B the compound being purified forms two layers with water
C the compound being purified has a lower boiling temperature than water
D the compound being purified is flammable. [1]

(Total for Question 2 = 1 mark)

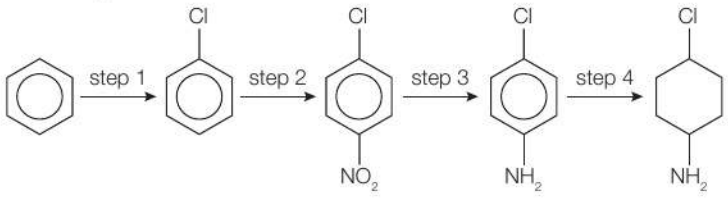
3 In the first stage of the synthesis of methyl 3-nitrobenzoate, methyl benzoate ($C_6H_5COOCH_3$) is prepared by the reaction of benzoic acid with methanol in the presence of concentrated sulfuric acid.

When the reaction is complete, the sulfuric acid is neutralised by adding aqueous sodium carbonate. What is the simplest way of obtaining the impure methyl benzoate from this mixture?

A refluxing
B solvent extraction
C filtration
D recrystallisation. [1]

(Total for Question 3 = 1 mark)

4 This question concerns the reaction scheme:



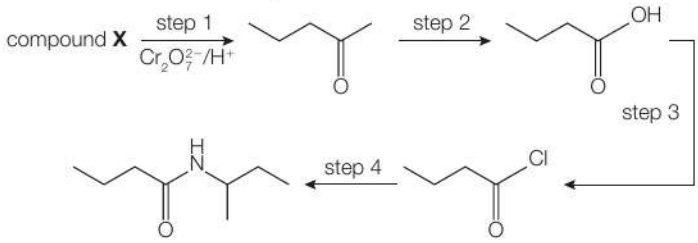
(a) Which step is most likely to need tin and concentrated hydrochloric acid?
A Step 1 **B** Step 2 **C** Step 3 **D** Step 4 [1]

(b) Which step is most likely to need a mixture of concentrated nitric acid and concentrated sulfuric acid?
A Step 1 **B** Step 2 **C** Step 3 **D** Step 4 [1]

(c) Which step is most likely to need a catalyst of aluminium chloride?
A Step 1 **B** Step 2 **C** Step 3 **D** Step 4 [1]

(Total for Question 4 = 3 marks)

5 Look at this reaction sequence:



(a) What is compound **X**?
A pentan-1-ol
B pentan-2-ol
C 2-methylbutan-1-ol
D 2-methylbutan-2-ol [1]

(b) Which substances are required for step 2?
A acidified potassium dichromate(VI)
B iodine in alkali, followed by hydrochloric acid
C sodium hydroxide solution followed by hydrochloric acid
D ammoniacal silver nitrate (Tollens' reagent) [1]

(c) Which is the reagent for step 3?
A aqueous chlorine
B chlorine gas
C hydrochloric acid
D phosphorus(V) chloride [1]

(d) Which is the reagent for step 4?
A $CH_3CH_2CH_2CH_2NH_2$
B $CH_3CH(NH_2)CH_2CH_3$
C $CH_3CH_2CH_2CONH_2$
D $CH_3CH(CONH_2)CH_3$ [1]

(Total for Question 5 = 4 marks)

6 Two organic compounds, **P** and **Q**, have the same molecular formula, C_4H_8O.

Both compounds contain a carbonyl group.

(a) Describe what you would see when 2,4-dinitrophenylhydrazine is added to either of these compounds. [1]

(b) Outline a test you could perform to show that **P** is a ketone and that **Q** is an aldehyde. [3]

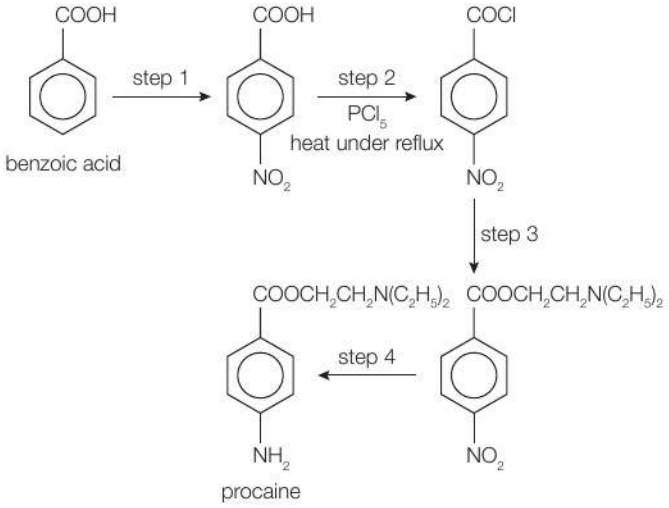
(c) (i) Give the structural formulae of the two possible isomers of **Q** that are aldehydes. [1]

(ii) Name the technique you would use to purify the product of the test with 2,4-dinitrophenylhydrazine. [1]

(iii) Other than by spectroscopic techniques, explain how you would use the purified product to identify compound **Q**. [2]

(Total for Question 6 = 8 marks)

7 Procaine can be synthesised from benzoic acid. The simplified reaction route is:



(a) Suggest the two reagents needed for step 1. [2]

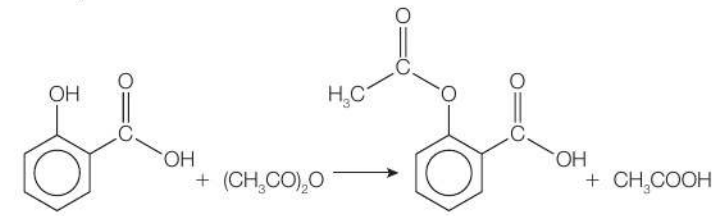
(b) Suggest why the reaction mixture in step 2 needs to be heated under reflux. [2]

(c) Give the structural formula for the organic reagent needed in step 3. [1]

(d) Name the type of reaction taking place in step 4. [1]

(Total for Question 7 = 6 marks)

8 A student produced a sample of aspirin by the esterification of 9.40 g of 2-hydroxybenzoic acid with excess ethanoic anhydride:



After purification by recrystallisation, 7.77 g of aspirin was obtained.

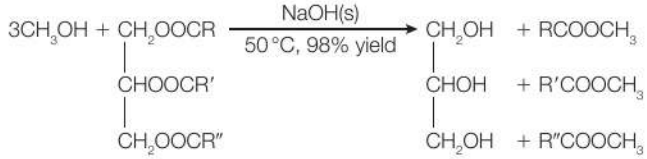
(a) Calculate the percentage yield obtained. [4]

(b) (i) Outline how you would purify a solid such as aspirin by recrystallisation, using water as the solvent. [5]

(ii) Explain what effect recrystallisation has on the final yield. [2]

(Total for Question 8 = 11 marks)

9 Biodiesel can be made from vegetable oils and methanol. The equation for the reaction is:

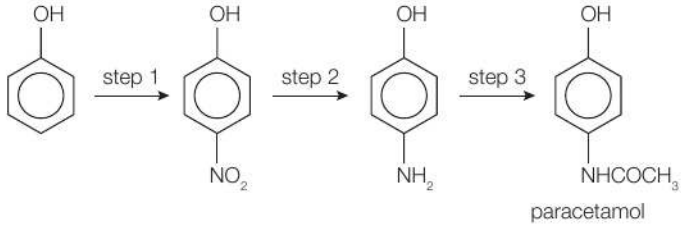


(a) Describe two hazards when carrying out this reaction. For each one, state the precaution you would take to minimise the risk. [4]

(b) Suggest two environmental benefits of carrying out this reaction. [2]

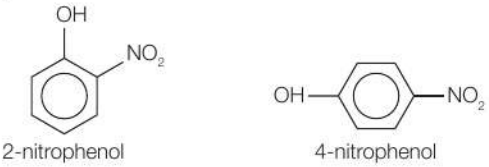
(Total for Question 9 = 6 marks)

10 Paracetamol is a widely used painkiller. It can be made from phenol as shown below.



(a) State the reagent(s) used in each of steps 1, 2 and 3. [3]

(b) In the nitration of phenol in step 1, two compounds are produced:



These compounds can be separated by steam distillation, since 2-nitrophenol is volatile in steam but 4-nitrophenol is not.

Describe briefly how 2-nitrophenol is obtained by steam distillation, and give one advantage of steam distillation over normal distillation. [4]

(c) The paracetamol obtained in the above synthesis can be purified by recrystallisation. In this process, the impure paracetamol is dissolved in the minimum volume of hot water and the mixture is then filtered while still hot. The filtrate obtained is cooled, and the resulting crystals are filtered and dried.

The table shows the solubility in water of paracetamol at various temperatures.

Temperature/°C	5	10	20	95
Solubility/g per 100 g	0.82	0.94	1.3	5.2

(i) Explain the purpose of each of the two filtrations in the recrystallisation process. [2]

(ii) Explain, using the temperatures given in the table, which pair of temperatures will give the highest yield of paracetamol. [2]

(iii) Name the technique that could be used in a school laboratory to check the purity of the recrystallised paracetamol. [1]

(Total for Question 10 = 12 marks)